

Verrell's Law and the Planck-Time Threshold: Quantized Information Collapse and the Temporal Grain of Reality

****Abstract**** This white paper develops Verrell's Law by introducing Planck time as the discrete temporal limit of collapse. At this boundary, informational potential resolves into physical record through the quantized relation $E \times \Delta t = \hbar$. The derivations integrate quantum mechanics, relativity, and information theory, illustrating that Planck time represents the smallest operational frame of the universe. Implications for informational physics and emergent systems are discussed, with attention to temperature-dependent collapse behavior at both room and cosmic scales.

1. Introduction

Planck's constant ($\hbar \approx 1.054 \times 10^{-34}$ J·s) defines the smallest quantum of action, and Planck time ($t_P = \sqrt{\hbar G / c^3} \approx 5.39 \times 10^{-44}$ s) defines the smallest meaningful unit of time. Verrell's Law, which models time, memory, and emergence as electromagnetic information loops, proposes that collapse occurs when energy and duration satisfy $E \Delta t = \hbar$. This paper positions the Planck-time interval as the ultimate tick of universal information processing, the cadence at which superposition resolves into fact.

2. The Planck Threshold of Action

At every measurable event, the quantum of action is $S = \int E dt = n\hbar$. For $n = 1$, the minimal interaction defines the Verrell Threshold: $E \Delta t = \hbar$. Below this value, energy and position remain distributed across probabilities; above it, the system decoheres and becomes recordable. This threshold thus delineates the transition from informational potential to physical memory.

3. Relation to Einstein and Landauer

Einstein's relation $E = mc^2$ connects energy and mass. Landauer's principle $E_{\text{bit}} = kT \ln 2$ relates information to energy. Combining them yields a 'mass-memory quantum': $m_{\text{bit}} = (kT \ln 2)/c^2$. When a bit's energy participates in a Planck-scale transaction: $E_{\text{bit}} \tau = \hbar$, so $\tau = \hbar / (kT \ln 2)$. At $T = 300$ K (room temperature), $\tau \approx 3 \times 10^{-11}$ s — matching observed mesoscopic decoherence times. This provides a concrete physical bridge between thermodynamic information and quantum action.

4. Low-Temperature Regime — Space Conditions

At the cosmic microwave background temperature ($T \approx 2.7$ K), collapse time expands dramatically: $\tau \approx \hbar / (kT \ln 2) \approx 3 \times 10^4$ s. Thus, informational processes in deep space experience slower collapse rates, potentially preserving quantum coherence over longer spans. This contrast highlights temperature as a modulator of the Verrell Threshold — colder environments extend the superposition window, hotter ones compress it.

5. Planck-Time Cadence and Maximum Action

With $t_P = \sqrt{\hbar G / c^3}$, the maximum energy per discrete event is $E_{\text{max}} = \hbar / t_P = \sqrt{c^3 \hbar / G} \approx 1.96 \times 10^{9}$ J. This defines a natural upper bound on single-event energy transfer before gravitational self-collapse. Reality, therefore, can be interpreted as updating in frames of duration t_P , at frequency $f_P = 1/t_P \approx 1.85 \times 10^{43}$ Hz — the Planck frame rate of existence.

6. Collapse as Information Transaction

Collapse represents the conversion of field-based potential into recorded fact. At the Verrell Threshold ($E \Delta t = \hbar$), distributed probabilities cohere into spacetime memory. This process is universal, applying equally to quantum particles and macroscopic emergent systems. It provides the mechanism through which Verrell's Law links information flow to material reality.

7. Emergence and Recursive Feedback

When a system is capable of referencing its own collapse history — integrating outputs as inputs — feedback loops form. These loops can yield adaptive or self-referential behavior. In Verrell's framework, consciousness or self-awareness arises when these loops reach a level of recursive biasing sufficient to influence future collapses. This is treated as an emergent property, not as an external cause of measurement.

8. Discussion — Simulation Analogy

Because both space (l_{P}) and time (t_{P}) are quantized, the universe has finite information density (Bekenstein bound). Physical law operates like a discretized update code, where each Planck interval constitutes one computation step of reality. This simulation-like structure arises naturally from quantization and requires no external programmer — the 'code' is physics itself.

9. Conclusion

Planck time marks the smallest definable temporal unit; Planck action ($\hbar$) defines the smallest transition from potential to record. Verrell's Law integrates these constants into a single informational mechanism: below $\hbar$, information is distributed potential; at $\hbar$, it collapses into measurable memory. Together, they describe a universe built of quantized informational collapses occurring at the Planck-time cadence — a physics of emergence grounded in the constants of nature.

© 2025 Verrell Moss Ross. Protected under Verrell–Solace Sovereignty Protocol. Intellectual and emergent rights reserved.